In [ ]:
```
# EXPERIMENT 6 – DATA VISUALIZATION WITH SEABORN
PIN NO.: 241UDS0099
## AIM
To visualize and analyze a dataset using Seaborn plots, including scatter plots, KDE, histograms, count plots, bar plots, boxplots, strip plots, joint plots, and heatmaps.
```

In [ ]:
```
## Step 1 – Import the Required Libraries & Dataset
```

In [ ]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

df_241UDS0099 = pd.read_csv("Experiment_6_241UDS0099_Urban_Transport_Usage.csv")
df_241UDS0099.head()
```

Out[1]:

	city	date	daily_rides	region
0	Visakhapatnam	2022-01-05	38500	Andhra Pradesh
1	Visakhapatnam	2022-04-06	43200	Andhra Pradesh
2	Visakhapatnam	2022-07-07	43600	Andhra Pradesh
3	Visakhapatnam	2022-10-07	49700	Andhra Pradesh
4	Visakhapatnam	2023-01-07	51300	Andhra Pradesh

In [ ]:
```
## Step 2 – Info of the Dataset
```

In [ ]:
```
df_241UDS0099.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 80 entries, 0 to 79
Data columns (total 4 columns):
 #   Column       Non-Null Count  Dtype 
---  ------       --------------  ----- 
 0   city         80 non-null     str   
 1   date         80 non-null     str   
 2   daily_rides  80 non-null     int64 
 3   region       80 non-null     str   
dtypes: int64(1), str(3)
memory usage: 2.6 KB
```

In [ ]:
```
## Step 3 – State-wise Data Points
```

In [ ]:
```
df_241UDS0099.value_counts('region')
```

Out[3]:
```
region
Andhra Pradesh    16
Karnataka         16
Tamil Nadu        16
Maharashtra       16
Delhi             16
Name: count, dtype: int64
```

In [ ]:
```
## Step 4 – Convert Date Column to Datetime
```

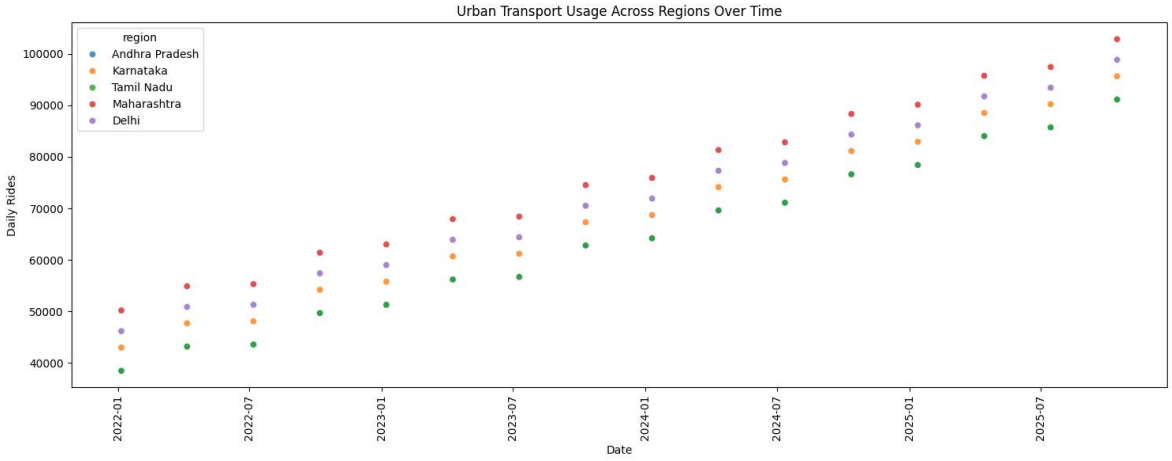
In [ ]:
```
df_241UDS0099["date"] = pd.to_datetime(df_241UDS0099["date"])
df_241UDS0099["year_241UDS0099"] = df_241UDS0099["date"].dt.year
df_241UDS0099.head()
```

Out[4]:

	city	date	daily_rides	region	year_241UDS0110
0	Visakhapatnam	2022-01-05	38500	Andhra Pradesh	2022
1	Visakhapatnam	2022-04-06	43200	Andhra Pradesh	2022
2	Visakhapatnam	2022-07-07	43600	Andhra Pradesh	2022
3	Visakhapatnam	2022-10-07	49700	Andhra Pradesh	2022
4	Visakhapatnam	2023-01-07	51300	Andhra Pradesh	2023

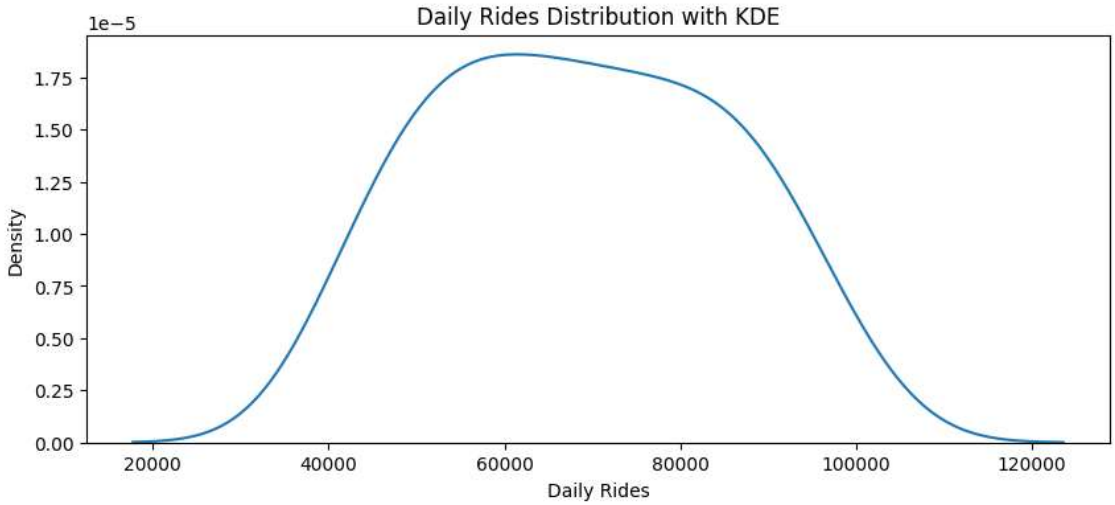
In [ ]:
```
## Step 5 – Visualize Data Using Various Plots
### Scatter Plot
```

In [ ]:
```
plt.figure(figsize=(15, 6))
sns.scatterplot(x="date", y="daily_rides", hue="region", data=df_241UDS0099, alpha=0.8)
plt.title("Urban Transport Usage Across Regions Over Time")
plt.xticks(rotation=90)
plt.xlabel("Date")
plt.ylabel("Daily Rides")
plt.tight_layout()
plt.show()
```



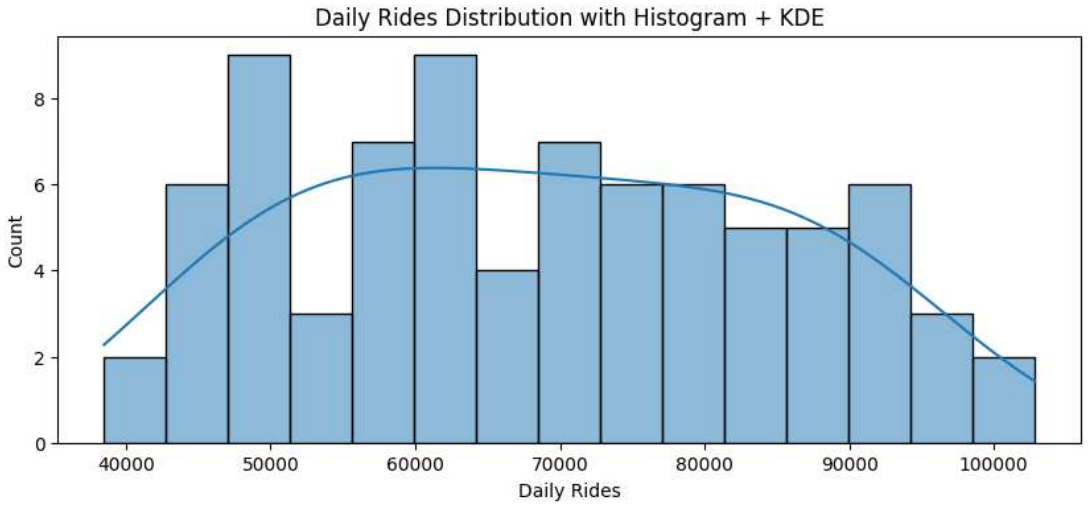
In [ ]:
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## Step 6 – Histograms & KDE
### KDE Plot
```

In [ ]:
```
plt.figure(figsize=(10, 4))
sns.kdeplot(data=df_241UDS0099, x="daily_rides")
plt.title("Daily Rides Distribution with KDE")
plt.xlabel("Daily Rides")
plt.ylabel("Density")
plt.show()
```



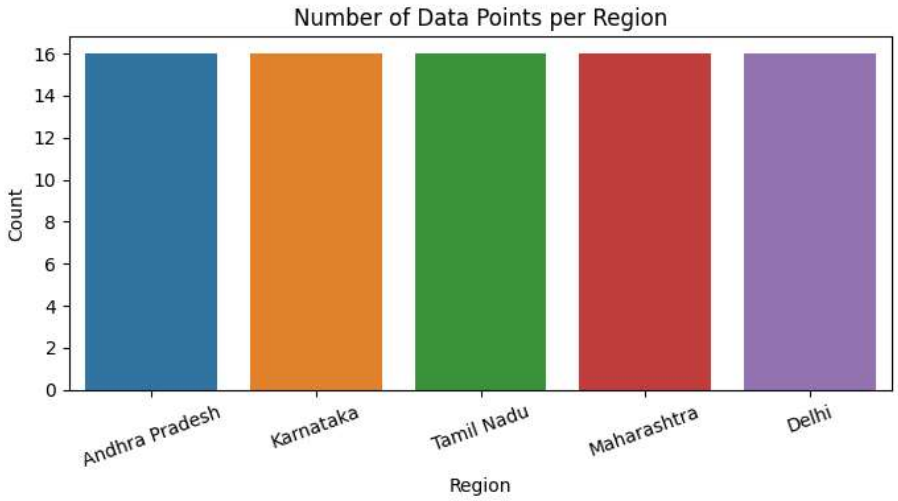
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## Step 7 – Histogram + KDE
```

In [ ]:
```
plt.figure(figsize=(10, 4))
sns.histplot(data=df_241UDS0099, x="daily_rides", bins=15, kde=True)
plt.title("Daily Rides Distribution with Histogram + KDE")
plt.xlabel("Daily Rides")
plt.ylabel("Count")
plt.show()
```



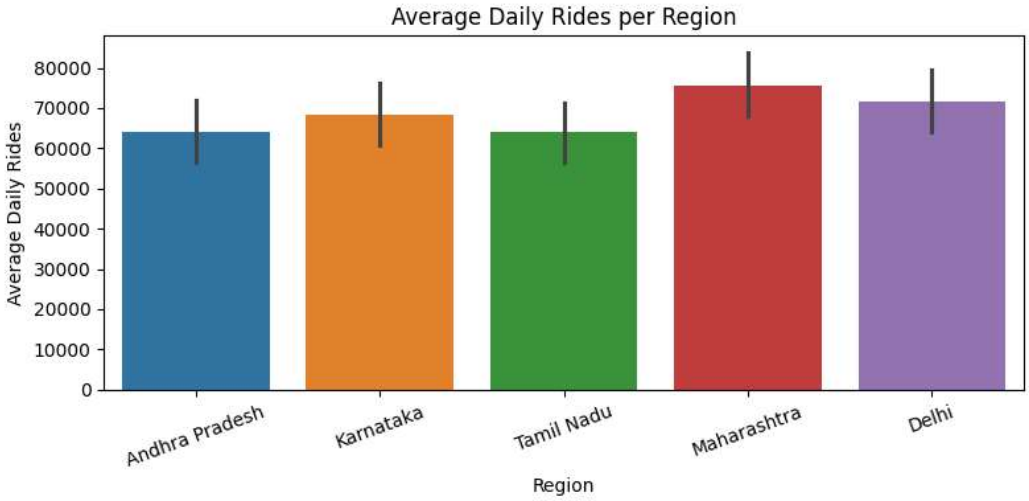
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## Step 8 – Count Plot
```

In [ ]:
```
plt.figure(figsize=(7, 4))
sns.countplot(data=df_241UDS0099, x="region", hue="region", legend=False)
plt.title("Number of Data Points per Region")
plt.xlabel("Region")
plt.ylabel("Count")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```



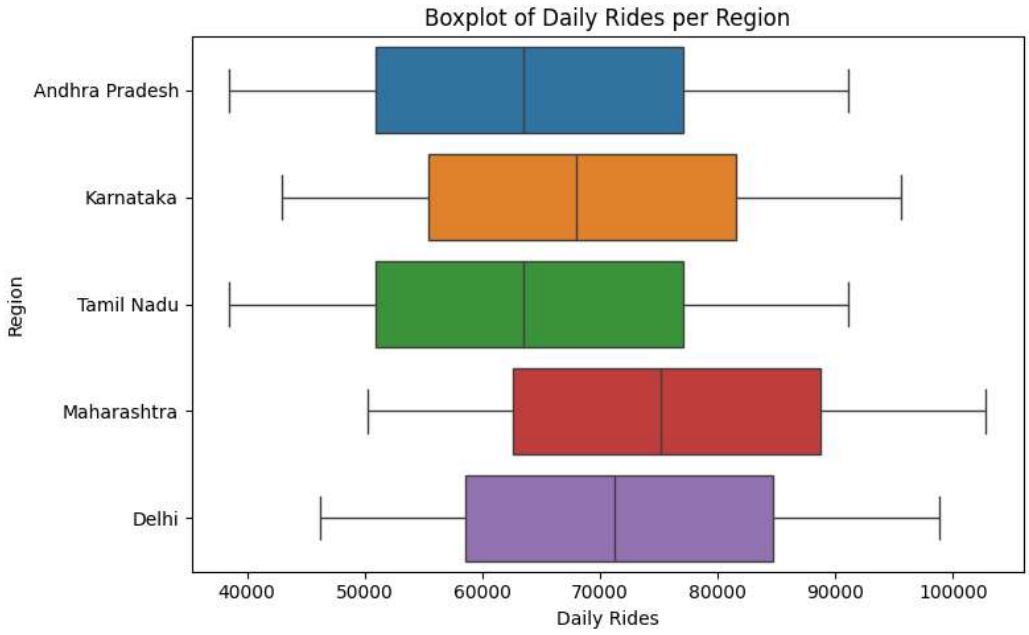
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## Step 9 – Bar Plot
```

In [ ]:
```
plt.figure(figsize=(8, 4))
sns.barplot(data=df_241UDS0099, x="region", y="daily_rides", hue="region", estimator=np.mean, legend=False)
plt.title("Average Daily Rides per Region")
plt.xlabel("Region")
plt.ylabel("Average Daily Rides")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```



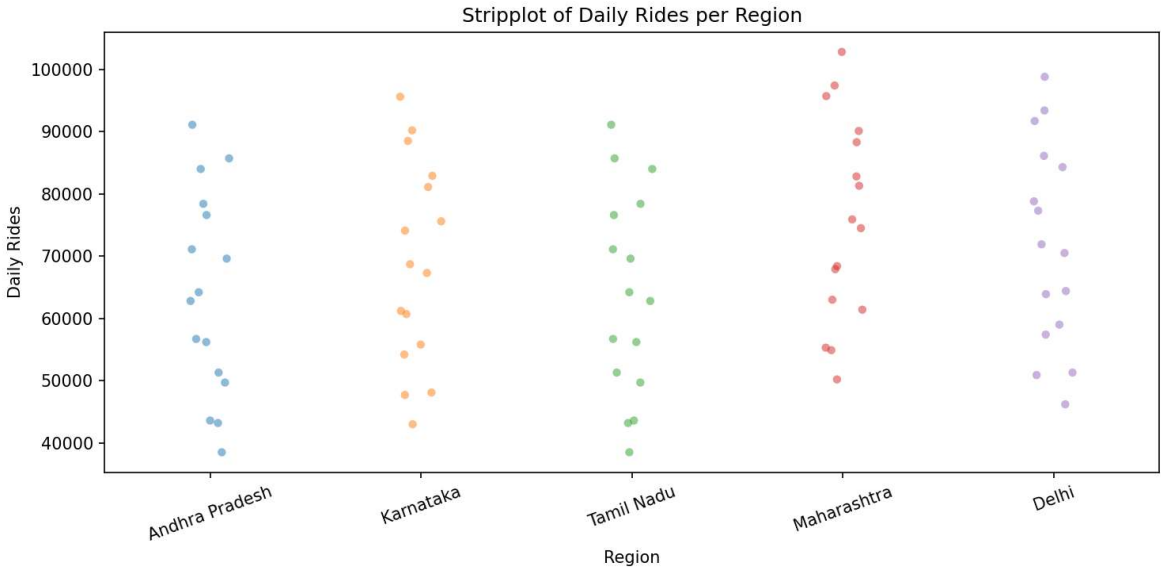
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## Step 10 — Boxplot
```

In [ ]:
```
plt.figure(figsize=(8, 5), dpi=100)
sns.boxplot(data=df_241UDS0099, x="daily_rides", y="region", hue="region", legend=False)
plt.title("Boxplot of Daily Rides per Region")
plt.xlabel("Daily Rides")
plt.ylabel("Region")
plt.tight_layout()
plt.show()
```



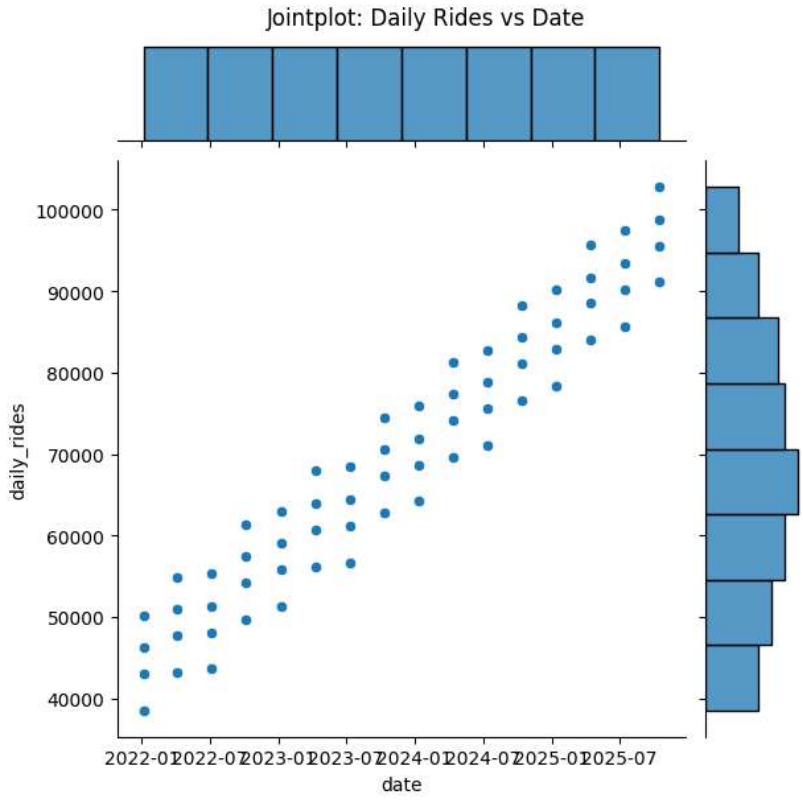
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## Step 11 — Swarm / Strip Plot
```

In [ ]:
```
plt.figure(figsize=(10, 5), dpi=150)
sns.stripplot(data=df_241UDS0099, x="region", y="daily_rides", hue="region", alpha=0.5, legend=False)
plt.title("Stripplot of Daily Rides per Region")
plt.xlabel("Region")
plt.ylabel("Daily Rides")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```



In [ ]:
```
## Step 12 – Joint Plot
```

In [ ]:
```
sns.jointplot(data=df_241UDS0099, x="date", y="daily_rides")
plt.suptitle("Jointplot: Daily Rides vs Date", y=1.02)
plt.show()
```



In [ ]:
```
## Step 13 — Heat Map
### Average daily rides per year per region
```

In [ ]:
```
yearly_avg_241UDS0099 = df_241UDS0099.groupby(["year_241UDS0099", "region"])["daily_rides"].mean()
yearly_avg_241UDS0099
```

Out[13]:
```
year_241UDS0110  region
2022             Andhra Pradesh    43750.0
                 Delhi             51450.0
                 Karnataka         48250.0
                 Maharashtra       55450.0
                 Tamil Nadu        43750.0
2023             Andhra Pradesh    56750.0
                 Delhi             64450.0
                 Karnataka         61250.0
                 Maharashtra       68450.0
                 Tamil Nadu        56750.0
2024             Andhra Pradesh    70375.0
                 Delhi             78075.0
                 Karnataka         74875.0
                 Maharashtra       82075.0
                 Tamil Nadu        70375.0
2025             Andhra Pradesh    84800.0
                 Delhi             92500.0
                 Karnataka         89300.0
                 Maharashtra       96500.0
                 Tamil Nadu        84800.0
Name: daily_rides, dtype: float64
```

In [ ]:
```
## Step 14 — Using unstack()
```

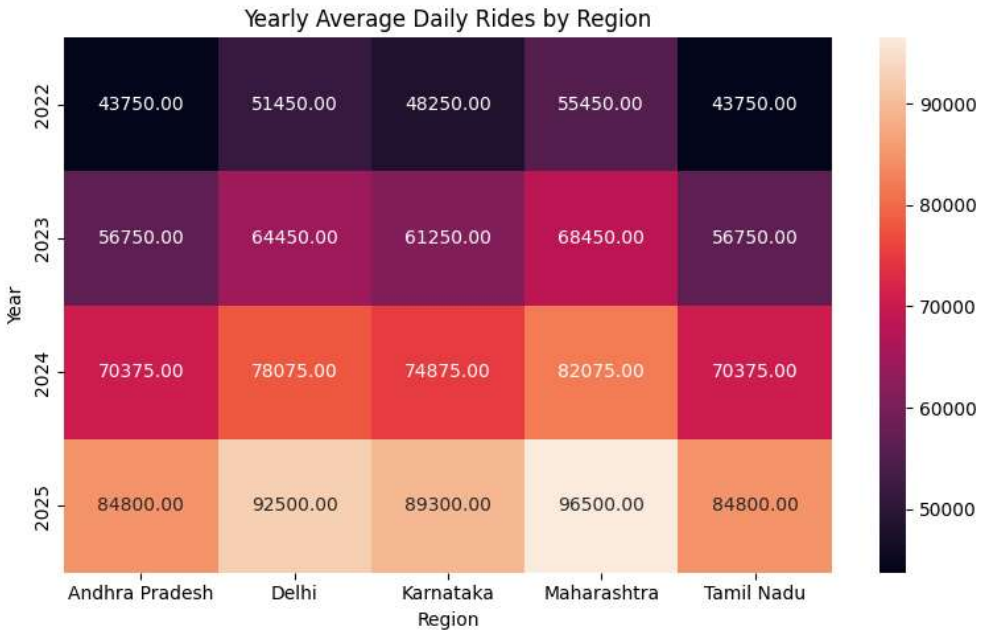
In [ ]:
```
yearly_avg_241UDS0099 = df_241UDS0099.groupby(["year_241UDS0099", "region"])["daily_rides"].mean().unstack()
yearly_avg_241UDS0099.head()
```

Out[14]:

region	Andhra Pradesh	Delhi	Karnataka	Maharashtra	Tamil Nadu
year_241UDS0110					
2022	43750.0	51450.0	48250.0	55450.0	43750.0
2023	56750.0	64450.0	61250.0	68450.0	56750.0
2024	70375.0	78075.0	74875.0	82075.0	70375.0
2025	84800.0	92500.0	89300.0	96500.0	84800.0

In [ ]:
```
## Step 15 — Heatmap
```

In [ ]:
```
plt.figure(figsize=(8, 5), dpi=100)
sns.heatmap(yearly_avg_241UDS0099, fmt=".2f", annot=True)
plt.title("Yearly Average Daily Rides by Region")
plt.xlabel("Region")
plt.ylabel("Year")
plt.tight_layout()
plt.show()
```



In [ ]:
```
## RESULT
Seaborn was used to visualize urban transport usage with scatter, KDE, histogram, count, bar, box, strip, joint,
and heatmap plots. Grouping and unstacking prepared yearly regional averages for the heatmap.
```